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Anomalous palaeomagnetic directions in a Mesozoic basaltic dyke; arguments for fault-induced remagnetization *

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Anomalously steep palaeomagnetic directions from the central part of an ~5 m wide basaltic dyke from Fossilryggen, East Antarctica, suggest a Recent to late Tertiary age for its remanent magnetization, in conflict with K–Ar isotope ages of 162 ± 4 and 217 ± 3 Ma obtained from the central and slickenside margins respectively. Three neighbouring basaltic intrusions carry stable magnetizations whose directions (mean D, 23°; mean I, −40° and pole position, 38° N, 40° E) accord with previously obtained Mesozoic results from lava flows in the Vestfjella basalt province, East Antarctica. Rock magnetic properties do not discriminate between the different dykes, and it is proposed that the anomalous directions represent spot-readings of the geomagnetic field which arise from complete remagnetization during a period of faulting in Recent to late Tertiary times.

1. Introduction

Vestfjella (73.74° S, 14.86° W) contains a number of small nunataks which in total cover less than 2% of an area of some 100 × 40 km, Fig. 1. The nunataks are formed from subhorizontal tholeiitic lava flows (1 to 10 m thick) cut by two suites of more or less vertical dykes striking NE–SW and NW–SE (Hjelle and Winsnes, 1972). The NE–SW group is the younger, and is associated with some 5% lateral extension of the region (Spaeth, 1986). The dykes are often associated with normal faults, downthrowing to the east as indicated by pronounced slickenside structures.

The lavas and dykes carry dominantly normal polarity magnetizations slightly steeper dipping than the reversed polarity basalts (Løvlie, 1988). Basalts carrying a reversed polarity magnetization, are all heavily overprinted with normal polarity components interpreted to be an expression of later thermochemical processes taking place during normal polarity periods of the geomagnetic field. This would imply that the reversely magnetized basalts represent the oldest magmatic phase in the Vestfjella region (Løvlie, 1988).

K–Ar isotopic age determinations of lavas and dykes from the region have yielded results ranging between 600 and 140 Ma (Furnes and Mitchell, 1978). The ages reported here are similarly inconsistent and are discussed in the context of geochemical, palaeomagnetic and structural factors.

At Fossilryggen, the nunataks comprise a group of some low-topography small exposures consisting of basaltic dykes cutting late Palaeozoic to early Mesozoic, continental to shallow marine sediments (Hjelle and Winsnes, 1972; Olaussen,

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1985). Anomalously steep-dipping, stable palaeomagnetic directions, compatible with a present to late Tertiary geomagnetic field configuration, have previously been reported from the central part and in the marginal slickenside zone of an ~ 5 m wide dyke (Løvlie, 1988). The inferred age of the remanent magnetization does not accord with the K–Ar isotopic ages of 162 ± 4 and 217 ± 3 Ma obtained from central and marginal samples. However, three adjacent basaltic intrusions carry normal polarity Mesozoic directions, suggesting that the anomalous dyke has suffered some local remagnetization process.

2. Geology and sampling

Because of the limited exposure of three of the sampled dykes at Fossilryggen (Fig. 1), it was not possible to unambiguously establish their modes of emplacement by investigating their margins and boundary zones. Pronounced slickenside structures were observed on the southeast side of an ~ 5 m wide, almost vertical, dyke indicating a downthrow to the south-east, which is in accordance with the general fault pattern in Vestfjella (Hjelle and Wisnes, 1972). This dyke strikes 040°–220°, suggesting that it belongs to the younger suite of dyke intrusions in Vestfjella. It is cut by several strike-parallel, calcite coated shear-planes. Oriented drill cores (19 mm in diameter) were collected with a water-cooled, two-stroke drill. Owing to weather conditions, orientation was by magnetic compass only. Sun-compass readings at other nunataks in the Vestfjella region indicate a magnetic deviation of −7°, with an accuracy of less than 2°.

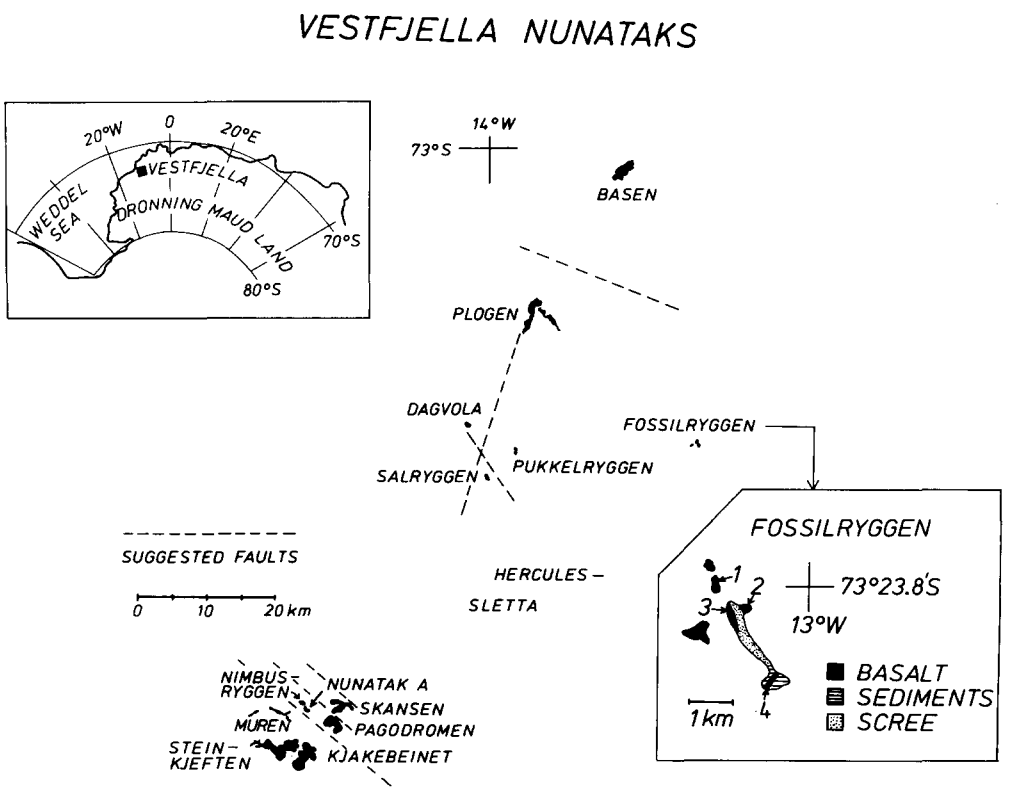


Fig. 1. Sketch map of the Vestfjella basaltic province and the Fossilryggen dykes and sampling localities, Dronning Maud Land, East Antarctica.

3. Magnetic properties

The Fossilryggen basalts are characterized by low natural remanent magnetization (NRM) intensities and by low bulk susceptibilities (except for dyke 2 which carries a significantly stronger remanence, see Table 1).

All samples were subjected to progressive thermal or alternating field (a.f.) demagnetization treatment. A Digico spinner was used for all measurements except for samples with NRM intensities below 2 mA m^{-1} which were measured on Cryogenic magnetometers at Newcastle upon Tyne (two-axes) and at Bergen (single axis inclined 45°).

Each dyke carries a stable, single component magnetization of normal polarity. Examples of directional behaviour during progressive demagnetization are shown in stereographic and optimum orthogonal vector projections in Fig. 2. In the latter projection, the horizontal plane is rotated along the best estimate of the declination. High magnetic stability is associated with low NRM intensities, and a.f. demagnetization to 80 mT was only partly successful in reducing NRM intensities by more than 75% (DM1A4, Fig. 2). A square-shaped thermal demagnetization decay curve (DM3A2) defines a narrow blocking temperature spectrum below 580 °C, indicative of an assembly

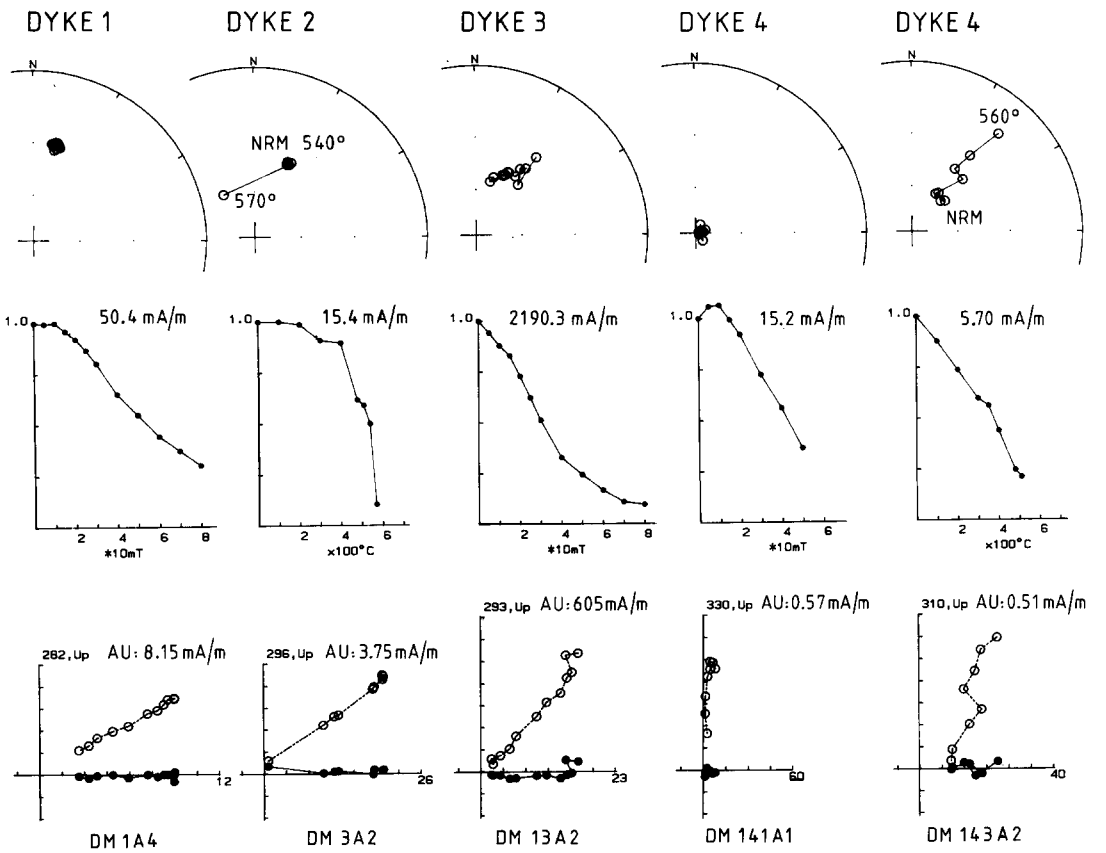


Fig. 2. Representative alternating field and thermal demagnetization results from the four dykes. The apparent relationship between low NRM intensities and high median destructive fields, and the square-shaped thermal decay curve for sample DM3A2 (dyke 2) defining a narrow blocking temperature interval below 580 °C should be noted. All stereographic directions are upper hemisphere. Orthogonal vector projections (lower diagrams): open and closed symbols correspond to vertical and horizontal projections respectively. AU is the axis unit.

TABLE 1

Magnetic properties of Fossilryggen dykes

Dyke	Sample	N	Intensity (mA m^{-1})	Susceptibility (SI 10^{-4})
1		5	115.1 ±172.7	15.5±19.6
2	Contact zone	5	4.22± 2.96	11.5± 0.4
3		3	1835 ±311	243.8±22.7
4	Centre	13	10.21± 8.07	5.0± 2.4
4	Margin	6	1.51± 0.34	5.4± 1.9

of magnetic grains of uniform magnetic size and composition.

Some samples exhibit elongated directional distributions (DM13A2, Fig. 2), reflecting the presence of multicomponent magnetizations which could not be resolved by the employed demagnetization treatment they were subjected to.

Mean declinations and inclinations, calculated by a line-find programme (Torsvik, 1987), coincide with the Mesozoic directions obtained from the Vestfjella lava flows lying further to the west (Løvlie, 1988) (see Table 2).

The central part of the 5 m wide dyke (dyke 4) carries anomalously steep-dipping directions (DM141A1 and DMN143A2, Fig. 2). During the initial demagnetization treatment, low-coercivity, reversed polarity components are removed in some samples (DM141A1, Fig. 2). These components do not define a distinct directional distribution, and they are therefore attributed to viscous components acquired after sampling. Thermal demagnetization (DM143A2, Fig. 2) was not successful above 500 °C owing to the acquisition of spurious viscous components probably reflecting the creation of superparamagnetic magnetite by the breakdown of titanomaghemite.

The slickenside samples have the lowest NRM intensities associated with the highest stabilities against a.f. demagnetization, and hence only partial demagnetization was obtained. Characteristic remanence directions define an elongate westerly oriented distribution, inconsistent with the distribution defined by the remanences of the adjacent dykes (Fig. 3).

TABLE 2

Palaeomagnetic results from Fossilryggen dykes and Vestfjella basaltic dykes and lavas

Dyke	N	*D*mean	*I*mean	α_{95}°	*k*
1	5	019	−36	8.0	92.8
2	5	018	−39	8.0	92.7
3	3	037	−45	13.2	88.0
Dykes 1–3	13	023	−40	5.4	59.9
4 centre	11	013	−78	4.6	97.4
4 margin	5	284	−63	11.9	42.1
[a] Vestfjella N	(22)	028	−57	4.1	55.5
[a] Vestfjella R	(5)	216	46	8.9	49.9

[a] Løvlie 1988: lavas and dykes combined.
N: number of samples
(N): number of sites
*D*mean: univectorial mean declination
*I*mean: univectorial mean inclination
α_{95}°: radius of 95% cone of confidence
k: precision parameter

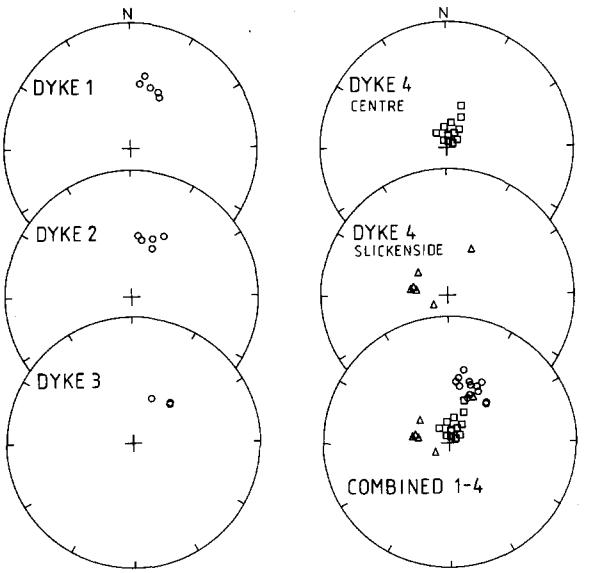


Fig. 3. Stereographic projections of characteristic remanence directions from dykes 1 to 4. It should be noted that some samples from both the central zone as well as from the slickenside margins of dyke 4 coincide with the general distribution from dykes 1–3.

4. Magnetomineralogy

4.1. Thermomagnetic analysis

Thermomagnetic curves were obtained from each sample on an automatic recording balance in

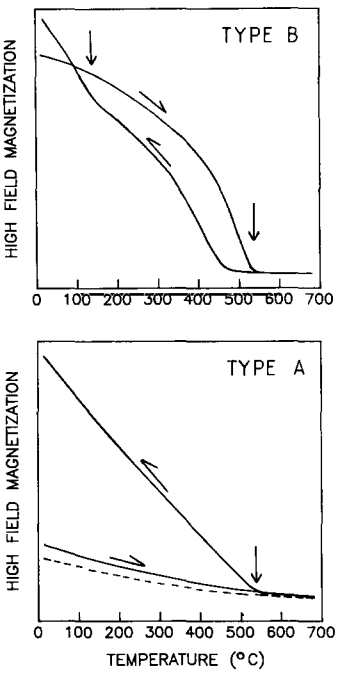


Fig. 4. Thermomagnetic curves from the Fossilryggen basalts. In curve type A the curves represent dykes 1, 2 and 4; the broken line represents the theoretical paramagnetic contribution calculated from the induced magnetization at 700 °C (973 K). Curve type B represents coarse-grained basalt (dyke 3). Curie temperatures are indicated by vertical arrows.

a 0.46 T field (the heating and cooling rate was 20 °C min^{-1} in air). Two distinctive types of irreversible thermomagnetic curves can be identified (Fig. 4). Type A is typical of samples having low NRM intensities (sites 1, 2 and 4); low initial high-field magnetizations decay monotonically exhibiting almost paramagnetic thermomagnetic characteristics with no indications of any unique Curie temperatures T_c (lower Fig. 4). Upon cooling from 700 °C, a strongly magnetic phase is created with a T_c between 520–570 °C, probably as a consequence of the inversion of titanomaghemites to almost pure magnetite (O'Reilly, 1983).

The type B curves only occur among samples from the coarse grained dyke 3; heating curves define a T_c between 520–540 °C, which is reduced to ~ 450 °C upon cooling from 700 °C (Fig. 4). In addition, cooling curves define a reversible T_c between 150–200 °C, which may arise from the creation of magnetic cristobalite, which is known to form from the thermal breakdown of chlorite (Moskowitz and Hargraves, 1984) (Fig. 4).

4.2. *Microscopy*

Polished sections of slices of cores taken from each site were examined in a reflected light microscope at magnifications up to ×1100.

All of the examined samples contained sulphide grains (mostly pyrite), homogeneous ilmenite and titanomagnetite grains, none of these minerals showing any features indicative of deuteric oxidation.

Dyke 2 contains homogeneous, euhedral (< 100 μm), grey-brown TM grains embedded in a fresh coloured matrix and shows no features indicative of low-temperature oxidative alteration.

The remaining dykes all comprise a brownish coloured matrix containing titanomagnetite grains showing different degrees of low-temperature oxidation features (granulation). The opaque mineralogy of the slickenside samples could not be determined owing to small grain sizes (< 5 μm).

4.3. *Magnetic fabric*

Magnetic fabric was determined from measurements of the anisotropy of magnetic susceptibility (AMS), measured in all samples using an induction bridge KLY-2 operating at 0.1 mT a.c. field.

The shape of the susceptibility ellipsoid is determined by the lengths of the principal axes (k_{max}, k_{int} and k_{min}), the ratios of which define the lineation $P1$ (k_{max}/k_{int}), anisotropy factor $P2$ (k_{max}/k_{min}), foliation $P3$ (k_{int}/k_{min}) and ellipticity E ($P3/P1$) (Collinson, 1983).

Isotropic magnetic minerals like magnetite or titanomagnetite have a shape-dependent anisotropy of magnetic susceptibility, expressing the spatial distributions of the principal geometric axes (Hrouda, 1982). Thus magnetic fabric may, in principle, be used to deduce the mode of emplacement of basalt. Empirical evidence (Ellwood, 1978) suggests that the distribution of the principal susceptibility axes in extrusive rocks is more scattered than is the case for intrusions, which often have well-defined distributions of k_{max} or k_{min} axes.

TABLE 3

Anisotropy of magnetic susceptibility parameters

Dyke	$P1$	$P2$	$P3$	E
1	1.005	1.008	1.003	0.999
	±0.005	±0.006	±0.002	±0.004
2	1.002	1.003	1.002	1.001
	±0.002	±0.001	±0.001	±0.001
3	1.010	1.028	1.018	1.008
	±0.005	±0.011	±0.010	±0.012
4 centre	1.003	1.006	1.003	1.000
	±0.003	±0.005	±0.002	±0.004
4 margin	1.005	1.005	1.003	1.000
	±0.001	±0.001	±0.001	±0.002

$P1$: lineation = k_{max}/k_{int}
$P2$: anisotropy factor = k_{max}/k_{min}
$P3$: foliation = k_{int}/k_{min}
E: ellipticity = $P3/P1$

The Fossilryggen basalts are characterized by low anisotropy factors ($P2$), generally below 1% (Table 3). The coarse-grained basalt from site 3 is an exception ($P2 = 1.8$–4.3%), this basalt also having the best-defined foliation plane (Fig. 5). Oblate ($E > 1$) and prolate ($E < 1$) susceptibility ellipsoids (indicative of compressional and tensional stresses respectively), are present within each site (the Flinn diagram, Fig. 5).

Distinct distributions of the principal susceptibility axes are clearly distinguishable in dykes 1, 2 and 4 (slickenside) (Fig. 5). Sites 1 and 3 are characterized by systematic distributions of the principal axes, defining magnetic fabric planes striking NW–SE and dipping eastwards in general agreement with the observed chilled margin surface at site 1. The actual strike of these dykes could not be determined in the field owing to snow cover. On the assumption that the magnetic fabric properties are related to the actual strike of the intrusions, we conclude that dykes 1 to 3 strike NW–SE. This suggests that they belong to the older suite of dykes intruding the Vestfjella lavas.

Although the principal axes determined for the central zone of dyke 4 are scattered, they nonetheless appear to relate to the observed strike of the dyke (040°–220°) (Fig. 5). The slickenside samples however, show a foliation plane which is coincident with the marginal faultplane (Fig. 5), demonstrating a close relationship between the spatial distribution of the titanomagnetite grain shapes and development of the slickensiding.

The distinctly different patterns of principal axes observed in dykes 1–3 compared with 4 are believed to be a consequence of the different mechanisms responsible for the observed magnetic fabrics. While dykes 1 to 3 probably retain their primary fabric, the fabric of dyke 4 is regarded as modified by faulting.

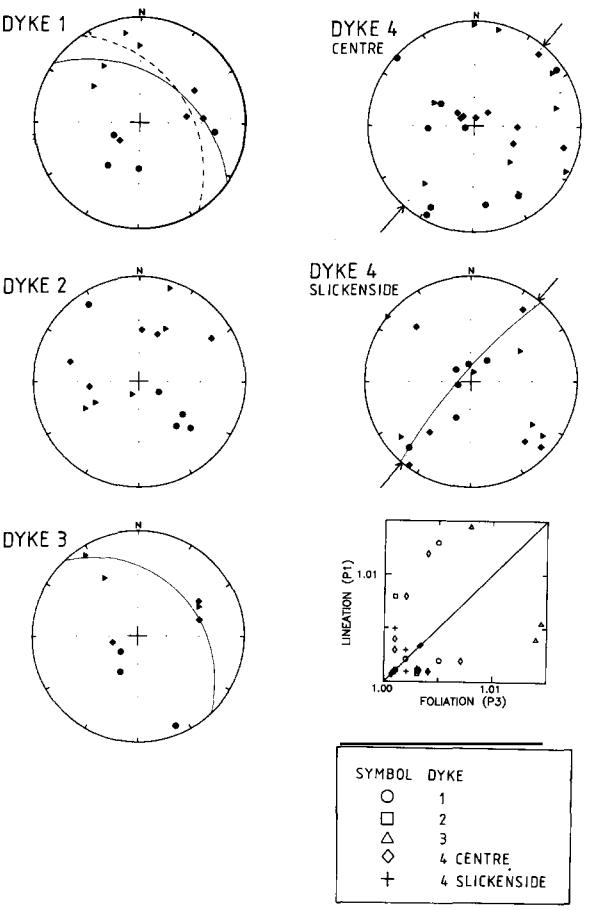


Fig. 5. Distributions of principal susceptibility axes and Flinn diagram from dykes 1–4. All directions lower hemisphere. Lower hemisphere foliation planes of magnetic fabric are indicated by great circles. Observed surface structure of dyke 1 is indicated by a broken line (lower hemisphere). Arrows indicate observed strike of dyke 4. Flinn diagram for all dykes at the bottom right.

5. Origin of magnetization

Stable, single-component magnetizations of normal polarity, in three of the basaltic dykes from the nunataks at Fossilryggen, define palaeomagnetic directions in general agreement with the Mesozoic palaeomagnetic field axes prevailing in the Vestfjella region (Table 2).

The inconsistent, steeply dipping magnetic directions in the adjacent dyke 4 may represent spot-readings of the present geomagnetic field configuration. However, this interpretation is not immediately compatible with the obtained Mesozoic K–Ar isotope ages (Table 4 and section 6). If dyke 4 is indeed Recent, then the apparent K–Ar ages would need to be explained via a mechanism of excess argon incorporation. Alternatively, the inconsistency between the age and remanence may arise from the dyke having been completely remagnetized. Considering the close lateral distance between the exposed dykes at Fossilryggen (less than 1000 m), any local Recent to late Tertiary magmatic event would have caused partial remagnetization of the adjacent dykes, which in fact carry a magnetization of demonstrably Mesozoic age. Since there is no evidence for steep, normal polarity overprinting in these intrusions, we are forced to conclude that the anomalous directions observed in dyke 4 cannot have been acquired at a time of dyke emplacement in Recent to late Tertiary time.

The Vestfjella region is cut by several major fault systems, and the inconsistent palaeomagnetic results from dyke 4 could be attributed to some local tectonic tilting. However, this dyke is observed to cut Permian continental to shallow marine sediments with distinctive bedding planes dipping only some 10° towards the NNW (Olaussen, 1985). The observed low angle bedding does not justify the application of any major correction to the palaeomagnetic directions, and it is concluded that dyke 4 has not experienced any major post-magnetization tilting.

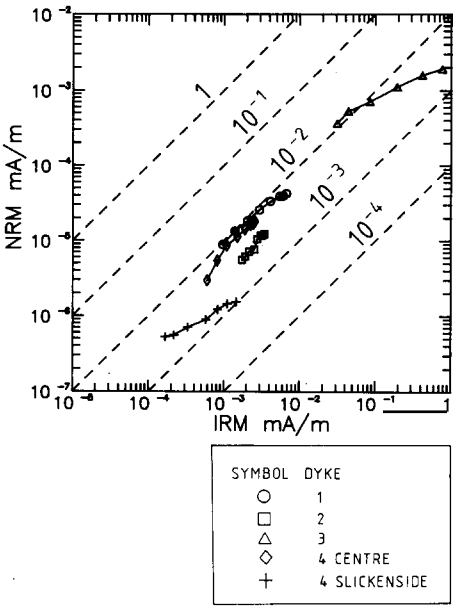


Fig. 6. Logarithmic plot of alternating field demagnetization of the natural remanent magnetization and saturation isothermal remanent magnetization for one sample from each dyke. Symbols as in Figure 5.

The elongated distribution of characteristic directions from the central zone of dyke 4, suggests the presence of composite, unresolved directions comprising both anomalously steep-dipping components as well as more shallow directions. These observations indicate that dyke 4 has experienced almost complete remagnetization in a steeply dipping field. This interpretation is also able to account for the existence of the single slickensided sample whose magnetization coincides with the group of Mesozoic directions identified in dykes 1 to 3 (Fig. 6).

The close grouping of the steep directions obtained from the centre of dyke 4 may be interpreted as indicating a remagnetizing process which acted over a long time interval. This interpretation is difficult to reconcile with the distinctly discor-

TABLE 4

Sample location in dyke 4	K_2O (wt.%)	Radiogenic ^{40}Ar ($mm^3\ g^{-1}$)	%Atmospheric contamination	Age (Ma) $\pm 1\sigma$
Margin	0.631	4.69×10^{-3}	38.9	217 ± 3
Centre	0.887	4.85×19^{-3}	28.0	162 ± 4

$\lambda_\varepsilon = 0.581\times10^{-10}\ a^{-1}$ $\lambda_\beta = 4.962\times10^{-10}\ a^{-1}$

$\frac{^{40}K}{K} = 1.167\times10^{-2}$ at%

dant and elongate distribution of the directions from the slickenside samples, suggesting spot-readings of the geomagnetic field.

We are led to conclude therefore, that the margin of dyke 4 was almost completely remagnetized at the time of faulting, and that this process also modified the magnetic fabric in the slickenside zone. The central part of the dyke has itself been subjected to a different episode of faulting (shearing?), evidenced by the calcite filled shear zones, responsible for almost complete remagnetization.

Since there is no rock magnetic evidence discriminating the anomalous dyke 4 from the remaining three dykes, any remagnetizing processes did not involve major chemical alterations. To elucidate any differences in the magnetization processes between the inferred primary magnetization carried by dykes 1–3 and the anomalous magnetization of dyke 4, the diagnostic method suggested by Fuller et al. (1988) has been applied. This method relies on the empirical observation that there is an apparent constant ratio between the thermoremanent magnetization (TRM) and saturation isothermal remanent magnetization (IRM(s)) of igneous rocks during progressive a.f. demagnetization (Fuller et al., 1988). For basaltic lavas, this ratio is between 10^{-1} and 10^{-2} for a primary TRM and significantly less for secondary magnetization (Fuller et al., 1988).

Progressive acquisition of IRM in samples from each dyke from Fossilryggen, revealed that complete saturation was achieved in fields around 0.4 T. Subsequent progressive a.f. demagnetization of the IRM(s) revealed almost constant ratios between NRM/IRM(s), reflected by almost straight lines in the logarithmic NRM–IRM(s) plot shown in Fig. 6. The ratios appear to increase with increasing alternating fields for both slickenside samples (dyke 4) as well as for samples from dyke 3. Thus this method does not discriminate between the inferred primary (deuteric) magnetization in dykes 1 to 3 and the anomalous magnetization in dyke 4.

The proposed origin of the anomalous palaeomagnetic directions retained in dyke 4 may be explained by complete remagnetization due to stress release associated with faulting in late Tertiary to Recent times. It has been demonstrated that igneous rocks exposed to uniaxial stress cycling may acquire a remanent magnetization parallel to the ambient field (Latham and Fuller, 1988). However, while the laboratory-produced stress-induced remanences were removed in alternating fields of 10 mT (Latham and Fuller, 1988), the inferred stress induced magnetizations residing in dyke 4, Fossilryggen, are retained beyond 60 mT fields. These conflicting observations may be accounted for by considering that dyke 4 was probably exposed to significantly higher stress ranges.

6. Isotopic age evidence

K–Ar ages were determined by the methods described by Wilkinson et al. (1986) using crushed portions of the core samples used in the magnetic studies.

Using the limited K–Ar data available (Table 4), it is possible to develop a model which establishes a lower stratigraphic boundary to the age of the evident recent palaeomagnetic (and therefore presumably geochemical) disturbance of dyke 4 arising from faulting. We assume that the age (162 ± 4 Ma) obtained from the interior of the dyke reflects its geological age. This age lies within the range of whole-rock K–Ar ages obtained from other basalt dykes in Vestfjella in Dronning Maud Land (Furnes and Mitchell, 1978) which were in the range 175 to 159 Ma (14 independent samples; 1σ uncertainties typical ± 3 Ma). The age obtained from the dyke margin (217 ± 3 Ma), we regard as having been perturbed. This increased age may be a consequence of the chilling of the dyke margin at the time of emplacement leading to the incorporation of the initial argon, and the finer grain size of the marginal sample would favour this explanation. However, no evidence of comparable levels of initial argon was detected in any of the previously dated basaltic dykes from Vestfjella which are of comparable size and similar petrology.

We prefer to interpret the older age from the margin as being a consequence of alteration in which fractional potassium loss has exceeded the

fractional argon loss. If a single, shortlived geological event (presumably that of faulting) was responsible for these losses, it would be possible to place a boundary on the time of the event, this being best demonstrated using an argon development diagram. For the purposes of the model, it is necessary to make the (unjustifiable) assumption that the initial potassium contents of the interior and marginal samples were identical.

In general, the radiogenic ^{40}Ar which accumulates in a sample in a time t is given by

$$^{40}Ar = \frac{\lambda_e}{\lambda} {}^{40}K(e^{\lambda t} - 1)$$

where ^{40}K corresponds to the present-day potassium content, λ_e is the electron capture decay constant and λ the total decay constant for ^{40}K. Since $\lambda = 5.5 \times 10^{-10}$ a^{-1}, for ages of the order of 10^8 a, this equation can be written to the first-order term by

$$^{40}Ar = \lambda_e {}^{40}K t$$

with an accuracy better than 3%. On this timescale, therefore, the growth of argon with time is approximately linear, having a slope determined by the potassium content.

The line T_1 A_1 in Fig. 7 shows the growth of ^{40}Ar for the sample taken from the interior of dyke 4, having a $K_2 = 0.887$ wt.%, and which has not been subjected to any perturbation. We assume the marginal sample evolved with the same potassium content until a time T_2 in the past along the line T_1X. At this time we assume the potassium content was reduced to its present-day value of 0.631 wt.%, without loss of argon, and thereafter to the argon content A_2 at the present day. The apparent K–Ar age of this perturbed sample (217 ± 3 Ma) is then the intersection of the extrapolated line XA_2 back to the time axis. The point of intersection of the lines XA_1 and XA_2 yields the age of the potassium loss event and may be calculated from the equations of the lines, or read from the diagram to be 26 ± 5 Ma.

If the possibility of argon loss is admitted, concomitant with potassium loss, then only a pathway such as T_1YZA_2 in the argon development diagram is consistent with the present day argon and potassium contents, the age of the loss event being T_3, where necessarily, $T_3 < T_2$. No pathway involving potassium loss prior to T_2 (such as T_1KLM, for example) is consistent with the present day abundances.

The argon isotopic data conforms, therefore to a model of single stage K–Ar loss occurring in the margin of dyke 4 during, or since, the Miocene.

7. Conclusion

The anomalous steeply dipping, single-component directions of magnetization residing in a

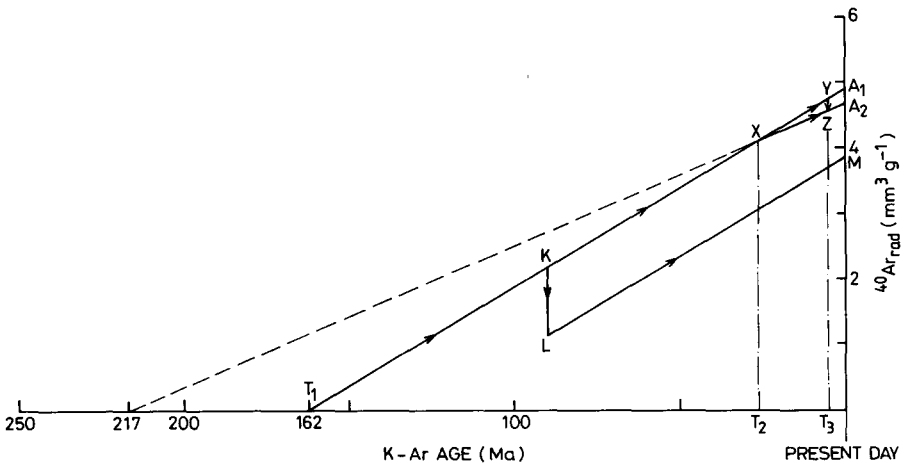


Fig. 7. Argon development diagram: model evolution of dyke 4.

basaltic dyke of Mesozoic age, is attributed to almost complete remagnetization as a consequence of episodes of late Tertiary to Recent faulting. The absence of significant differences in rock magnetic properties between the remagnetized dyke and three adjacent dykes carrying presumed primary Mesozoic palaeomagnetic directions and the non-conclusive results of the NRM: IRM(s) diagnostic method (Fuller et al., 1988), suggest that the inferred tectonically acquired magnetization did not significantly change the bulk magnetic properties. However, the magnetic fabric style of the remagnetized dyke is distinctly different from that of the adjacent dykes, implying that magnetic fabric has the potential to reflect remagnetization caused by tectonic activity. The limited K–Ar age data is consistent with a model of episodic potassium and argon loss occurring in the margin of the dyke in post-Miocene times, in accordance with the palaeomagnetic evidence of the time of thermochemical perturbation of the dyke.

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